Seminar 2:

Classroom problems and worked solutions

25117 - Econometrics

Universitat Pompeu Fabra

October 19, 2026

The issue: evidence about student and school performance

- We want to assess claims about average test scores and differences between groups.
- Sampling uncertainty determines how precisely we can estimate a mean and how much evidence a difference provides.
- For each problem, write the formula, show the calculation, and state what the result tells us about the population.

Classroom problems

Work through each problem before viewing its solution. Use the assumptions and normal probabilities supplied in the question.

Problem 1: How precise is the average test score?

Test scores $Y_1, \ldots, Y_n$ are i.i.d. with population standard deviation $\sigma_Y = 10$ points. Initially, $n = 100$.

1. Find $\mathrm{Var}(\overline{Y})$ and the theoretical standard deviation of $\overline{Y}$.
2. Now suppose σ_Y is unknown and the sample standard deviation is $s_Y = 10$ points. Estimate the standard error of $\overline{Y}$. How does it differ from the standard deviation of individual scores?
3. Repeat both calculations for $n = 400$. For this comparison, keep the population standard deviation and the supplied sample standard deviation at 10 points. What happens to precision when the sample size quadruples?

Solution 1: Individual dispersion and uncertainty in an average

1. For i.i.d. scores, the variance of the mean and its square root are

$$\mathrm{Var}(\overline{Y}) = \frac{\sigma_Y^2}{n} = \frac{10^2}{100} = 1 \text{ point}^2, \qquad \mathrm{SD}(\overline{Y}) = \sqrt{\frac{\sigma_Y^2}{n}} = \sqrt{1} = 1 \text{ point}.$$

2. When σ_Y is unknown, replace it with s_Y:

$$SE(\overline{Y}) = \sqrt{\frac{s_Y^2}{n}} = \frac{s_Y}{\sqrt{n}} = \frac{10}{\sqrt{100}} = \boxed{1 \text{ point}}.$$

The supplied individual-score SD is 10 points; the estimated uncertainty in the sample mean is 1 point.

3. With $n = 400$, holding both supplied SDs at 10 points,

$$\mathrm{Var}(\overline{Y}) = \frac{\sigma_Y^2}{n} = \frac{100}{400} = .25 \text{ points}^2,$$

$$\mathrm{SD}(\overline{Y}) = \sqrt{\frac{\sigma_Y^2}{n}} = \sqrt{.25} = .50 \text{ points},$$

$$SE(\overline{Y}) = \sqrt{\frac{s_Y^2}{n}} = \sqrt{\frac{100}{400}} = .50 \text{ points}.$$

Quadrupling n quarters the variance and halves the SD and estimated SE.

Problem 2: Is the average score different from 70?

An i.i.d. sample of 100 students has mean score $\bar{y} = 72.4$ and sample standard deviation $s_Y = 10$ points. Use the large-sample normal approximation.

1. State the null and alternative hypotheses for a two-sided test of whether the population mean is 70 points.
2. Calculate the estimated standard error, the observed t-statistic, and the two-sided p-value. You may use $\Phi(2.4) = .9918$, where Φ is the standard normal c.d.f.
3. Decide whether to reject at the 10%, 5%, and 1% levels. Check your decisions using the corresponding two-sided cutoffs 1.645, 1.96, and 2.576.
4. Interpret the p-value conditional on the null hypothesis. Does it give the probability that the null hypothesis is true?

Solution 2: Hypotheses, standard error, and the p-value

1. We test the population mean μ_Y:

$$H_0 : \mu_Y = 70 \qquad \text{against} \qquad H_1 : \mu_Y \neq 70.$$

2. The estimated standard error is the square root of the estimated sampling variance:

$$SE(\overline{Y}) = \sqrt{\frac{s_Y^2}{n}} = \frac{s_Y}{\sqrt{n}} = \frac{10}{\sqrt{100}} = 1 \text{ point.}$$

$$t^{\text{obs}} = \frac{\overline{y} - \mu_Y^0}{SE(\overline{Y})} = \frac{72.4 - 70}{1} = \boxed{2.4}.$$

Under H_0, the large-sample approximation gives $T \approx \mathcal{N}(0, 1)$. Thus

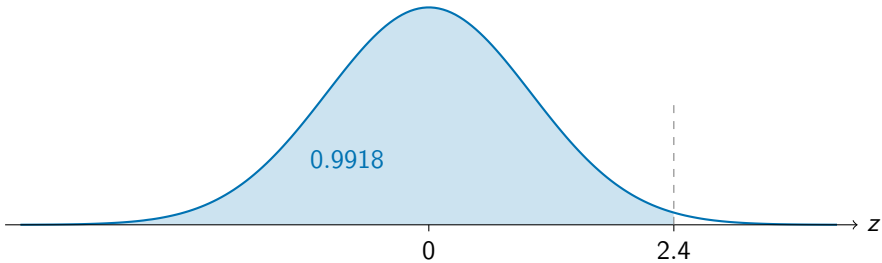
$$\text{p-value} \approx 2[1 - \Phi(|t^{\text{obs}}|)] = 2[1 - .9918] = \boxed{.0164}.$$

Both tails count because H_1 allows means above or below 70.

Solution 2: Step 1 — Read the cumulative probability

The standard normal CDF $\Phi(2.4)$ gives the area **to the left of** 2.4:

$$|t^{\text{obs}}| = 2.4, \text{ and } \Phi(2.4) = 0.9918.$$

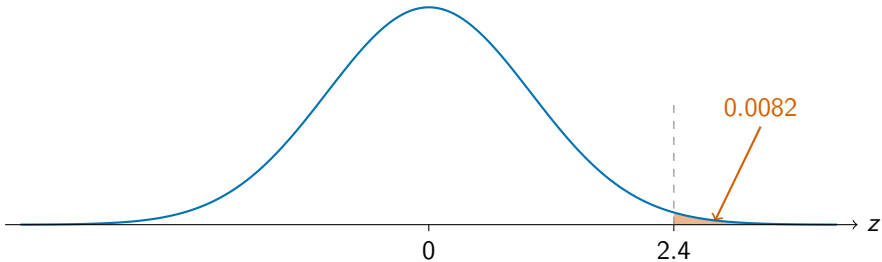


The blue area is 99.18% of the total. The whole area under the curve equals 1.

Solution 2: Step 2 — Keep the right tail

Subtract the area to the left from 1 to obtain the area **to the right of** 2.4:

$$1 - \Phi(|t^{\text{obs}}|) = 1 - \Phi(2.4) = 1 - 0.9918 = 0.0082.$$

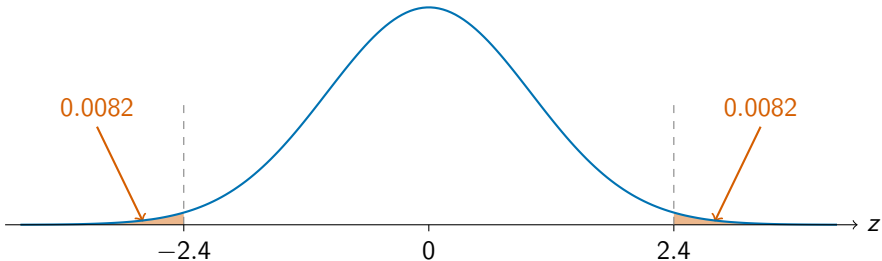


The red area is 0.82% of the total. This counts only the right tail.

Solution 2: Step 3 — Count both tails

The two-sided alternative counts equally extreme values: **below** -2.4 **or above** 2.4.

$$\text{p-value} \approx 2[1 - \Phi(|t^{\text{obs}}|)] = 2 \times 0.0082 = \boxed{0.0164}.$$



By symmetry, each tail has probability 0.0082. Together they give 1.64%.

Solution 2: Decisions and the meaning of .0164 (continued)

3. Reject H_0 when the p-value is at most α, or equivalently when $|t^{\text{obs}}|$ reaches the two-sided critical value.

Level α	Critical value	Compare $\lvert t^{\text{obs}}\rvert = 2.4$	Decision
10%	1.645	$2.4 > 1.645$	Reject: $.0164 < .10$
5%	1.960	$2.4 > 1.960$	Reject: $.0164 < .05$
1%	2.576	$2.4 < 2.576$	Fail to reject: $.0164 > .01$

4. If the population mean were 70, the probability of a test statistic at least as far from zero as 2.4, in either direction, would be approximately **1.64%** under the large-sample approximation.

What is being conditioned on?

The calculation assumes H_0 to obtain a probability of data as extreme as the observed result. It does not calculate the probability that H_0 is true after observing the data.

Problem 3: Does a larger sample change the evidence?

Two independent surveys draw i.i.d. test scores from the same student population:

Survey	Sample size n	Mean $\bar{y}$	Sample SD s_Y
A	100	71.5	10
B	400	71.5	10

In each survey, test $H_0 : \mu_Y = 70$ against $H_1 : \mu_Y \neq 70$ using the large-sample normal approximation.

1. Calculate each survey's estimated standard error and observed t-statistic.
2. Find each two-sided p-value. Use $\Phi(1.5) = .9332$ and $\Phi(3) = .99865$.
3. Give the 5% decision for each survey. Explain why the evidence differs even though both observed mean scores are 71.5.

Solution 3: The same difference measured with greater precision

1. For each survey, first take the square root of the estimated variance:

$$SE(\overline{Y}) = \sqrt{\frac{s_Y^2}{n}} = \frac{s_Y}{\sqrt{n}}, \qquad t^{\text{obs}} = \frac{\overline{y} - \mu_Y^0}{SE(\overline{Y})}.$$

Survey	Estimated SE (points)	Observed t-statistic
A	$10/\sqrt{100} = 1$	$(71.5 - 70)/1 = 1.5$
B	$10/\sqrt{400} = .5$	$(71.5 - 70)/.5 = 3$

2. Using p-value $\approx 2[1 - \Phi(|t^{\text{obs}}|)]$,

$$p_A \approx 2(1 - .9332) = \boxed{.1336}, \qquad p_B \approx 2(1 - .99865) = \boxed{.0027}.$$

3. At 5%, A **fails to reject** ($.1336 > .05$); B **rejects** ($.0027 < .05$).

Both estimate a difference of 1.5 points. With the same supplied sample SD, quadrupling the sample size halves the SE, so the observed t-statistic doubles.

Problem 4: Which testing mistakes can we make?

Scores are i.i.d. normal with mean μ and known standard deviation $\sigma = 10$. With $n = 100$, test

$$H_0 : \mu = 70 \qquad \text{against} \qquad H_1 : \mu \neq 70.$$

Reject if $|Z| \geq 1.96$, where $Z = (\overline{Y} - 70)/1$.

1. Find the sample means for which we do not reject. Compute the Type I error probability when $\mu = 70$.
2. If the true mean is $\mu = 73$, compute the Type II error probability β and the power of this test.
3. In 1,000 repetitions, how many Type I errors do we expect when $\mu = 70$? How many detections do we expect when $\mu = 73$?

Use $\Phi(1.96) = .9750$, $\Phi(-1.04) = .1492$, and $\Phi(-4.96) \approx 0$.

Solution 4: Rejection region and Type I error

1. The sampling standard deviation of the mean is

$$SE(\overline{Y}) = \frac{\sigma}{\sqrt{n}} = \sqrt{\frac{\sigma^2}{n}} = \sqrt{\frac{100}{100}} = \sqrt{1} = 1 \text{ point.}$$

We do not reject when

$$\left|\frac{\overline{Y} - 70}{1}\right| < 1.96 \quad \Longleftrightarrow \quad \boxed{68.04 < \overline{Y} < 71.96}.$$

When $\mu = 70$, $Z \sim \mathcal{N}(0, 1)$, so

$$\begin{aligned} \alpha &= \Pr(\text{reject } H_0 \mid \mu = 70) \\ &= \Pr(|Z| \geq 1.96) = 2[1 - \Phi(1.96)] \\ &= 2(1 - .9750) = \boxed{.05}. \end{aligned}$$

A Type I error means rejecting the null when the null is true.

Solution 4: Type II error and power (continued)

2. When the true mean is 73, $\overline{Y} \sim \mathcal{N}(73, 1)$, with variance 1. Hence

$$\begin{aligned}\beta(73) &= \Pr(\text{do not reject } H_0 \mid \mu = 73) \\ &= \Pr(68.04 < \overline{Y} < 71.96 \mid \mu = 73) \\ &= \Phi\left(\frac{71.96 - 73}{1}\right) - \Phi\left(\frac{68.04 - 73}{1}\right) \\ &= \Phi(-1.04) - \Phi(-4.96) \approx \boxed{.1492}.\end{aligned}$$

$$\text{Power}(73) = 1 - \beta(73) \approx 1 - .1492 = \boxed{.8508}.$$

3. Expected counts equal repetitions times the relevant probability:

$$\begin{aligned}\mu = 70: &\quad 1{,}000\alpha = 1{,}000(.05) = \boxed{50} \text{ Type I errors,} \\ \mu = 73: &\quad 1{,}000[1 - \beta(73)] \approx 850.8 \approx \boxed{851} \text{ detections.}\end{aligned}$$

These are expectations in two different settings, not guaranteed counts.

Problem 5: Which mean scores remain plausible?

A large i.i.d. sample of $n = 100$ students gives $\bar{y} = 72.4$ and sample standard deviation $s_Y = 10$ points. Use the normal approximation.

1. Calculate the 90%, 95%, and 99% confidence intervals for μ. Use critical values 1.645, 1.96, and 2.576, respectively.
2. Use the 95% interval to test each null against a two-sided alternative at the 5% level: $H_0 : \mu = 70$ and $H_0 : \mu = 73$. Explain the connection between the interval and the tests.
3. Interpret 95% coverage. Across 200 independently repeated samples, how many 95% intervals would you expect to contain the true mean?

Solution 5: Confidence levels and interval width

1. First compute the estimated standard error:

$$SE(\overline{Y}) = \frac{s_Y}{\sqrt{n}} = \sqrt{\frac{s_Y^2}{n}} = \sqrt{\frac{100}{100}} = \sqrt{1} = 1 \text{ point.}$$

The normal approximate interval is

$$CI_{1-\alpha} = \left[\bar{y} - z_{1-\alpha/2} SE(\overline{Y}),\ \bar{y} + z_{1-\alpha/2} SE(\overline{Y})\right].$$

Confidence	Calculation	Interval (points)
90%	$72.4 \pm 1.645(1)$	[70.755, 74.045]
95%	$72.4 \pm 1.96(1)$	[70.44, 74.36]
99%	$72.4 \pm 2.576(1)$	[69.824, 74.976]

Greater coverage requires a wider interval for the same data.

Solution 5: Test inversion and repeated coverage (continued)

2. A two-sided 5% test rejects μ_0 when

$$\left|\frac{\bar{y}-\mu_0}{SE(\bar{Y})}\right| > 1.96 \quad \Longleftrightarrow \quad \mu_0 \notin [\bar{y} - 1.96SE,\ \bar{y} + 1.96SE].$$

Thus the 95% interval contains the null values that are not rejected:

$$\mu_0 = 70: \quad t = (72.4 - 70)/1 = 2.4, \quad \boxed{\text{reject } H_0},$$
$$\mu_0 = 73: \quad t = (72.4 - 73)/1 = -.6, \quad \boxed{\text{do not reject } H_0}.$$

3. For intervals constructed by this method,

$$\Pr\big(\mu \in [\bar{Y} - 1.96SE(\bar{Y}),\ \bar{Y} + 1.96SE(\bar{Y})]\big) \approx .95.$$

The mean μ is fixed; the endpoints vary across samples. Approximately 95% of intervals cover μ in repeated sampling. Among 200 repetitions, the expected number covering μ is

$$200(.95) = \boxed{190}.$$

Problem 6: How many students give the desired precision?

Plan an i.i.d. survey of scores using a preliminary standard deviation of $\sigma = 10$ points. Use a 95% normal approximate confidence interval.

1. Derive the minimum sample size needed for an interval half-width of at most 1 point. Round up to a whole student.
2. How large must the sample be for a half-width of at most .5 points? Explain the relationship between precision and sample size.
3. If the planning standard deviation is 20 points instead, how large a sample is needed for a half-width of at most 1 point? Explain the role of the planning assumption.

Solution 6: Sample size for a target half-width

1. The planning standard error and interval half-width are

$$SE(\overline{Y}) = \sqrt{\frac{\sigma^2}{n}} = \frac{\sigma}{\sqrt{n}}, \qquad m_n = 1.96\frac{\sigma}{\sqrt{n}}.$$

For $m_n \le m$, solve and round up:

$$n \ge \left(\frac{1.96\sigma}{m}\right)^2 \quad \Longrightarrow \quad n_{\min} = \left\lceil \left(\frac{1.96\sigma}{m}\right)^2 \right\rceil.$$

$$\sigma = 10,\ m = 1: \quad n_{\min} = \lceil (19.6)^2 \rceil = \lceil 384.16 \rceil = \boxed{385}.$$

2. For $\sigma = 10$ and $m = .5$,

$$n_{\min} = \lceil (1.96 \cdot 10/.5)^2 \rceil = \lceil 1536.64 \rceil = \boxed{1537}.$$

Halving the target half-width multiplies required n by four before rounding.

3. For $\sigma = 20$ and $m = 1$, $n_{\min} = \lceil (39.2)^2 \rceil = \boxed{1537}$. Doubling σ also multiplies required n by four. Realized precision depends on how well the planning value describes the population.

Problem 7: Do two groups have the same average score?

Two independent i.i.d. samples give the following summaries. Use the large-sample normal approximation and let $\delta = \mu_A - \mu_B$.

Group	n	$\bar{y}$	s
A	100	75	12
B	144	71	12

1. Estimate δ and calculate its standard error.
2. Test $H_0 : \delta = 0$ against $H_1 : \delta \neq 0$. Find the test statistic and p-value. Decide at the 5% and 1% levels.
3. Construct a 95% confidence interval for δ.
4. Test $H_0 : \delta = 3$ against $H_1 : \delta \neq 3$ at 5%. Check your decision using the confidence interval.

Use $\Phi(2.561) \approx .99478$ and $\Phi(.640) \approx .7389$.

Solution 7: Difference, standard error, and a zero-null test

1. The estimated difference is $\widehat{\delta} = \overline{y}_A - \overline{y}_B = 75 - 71 = \boxed{4}$ points. Independence makes the covariance of the sample means zero:

$$\text{Var}(\overline{Y}_A - \overline{Y}_B) = \frac{\sigma_A^2}{n_A} + \frac{\sigma_B^2}{n_B}.$$

Estimate the variance, then take its square root:

$$SE(\widehat{\delta}) = \sqrt{\frac{s_A^2}{n_A} + \frac{s_B^2}{n_B}} = \sqrt{\frac{144}{100} + \frac{144}{144}} = \sqrt{2.44} \approx \boxed{1.562} \text{ points.}$$

2. For a hypothesized difference d_0,

$$t = \frac{\widehat{\delta} - d_0}{SE(\widehat{\delta})}, \qquad p = 2[1 - \Phi(|t|)].$$

With $d_0 = 0$, $t = 4/\sqrt{2.44} \approx 2.561$, and

$$p \approx 2(1 - .99478) = \boxed{.01044}.$$

Reject at 5% because $.01044 < .05$; do not reject at 1% because $.01044 > .01$.

Solution 7: An interval and a nonzero null (continued)

3. The 95% normal approximate confidence interval is

$$\begin{aligned} CI_{.95}(\delta) &= [\widehat{\delta} - 1.96SE(\widehat{\delta}),\ \widehat{\delta} + 1.96SE(\widehat{\delta})] \\ &= [4 - 1.96\sqrt{2.44},\ 4 + 1.96\sqrt{2.44}] \\ &\approx \boxed{[.938,\ 7.062]} \text{ points.} \end{aligned}$$

4. Changing the null value to $d_0 = 3$ changes the numerator:

$$t = \frac{\widehat{\delta} - 3}{SE(\widehat{\delta})} = \frac{4 - 3}{\sqrt{2.44}} \approx .640.$$

$$p = 2[1 - \Phi(|t|)] \approx 2(1 - .7389) = \boxed{.5222}.$$

Do not reject $\delta = 3$ at 5%. Consistently, 3 lies inside the 95% interval.

Subtracting a constant d_0 does not change the variance: $\text{Var}(\widehat{\delta} - d_0) = \text{Var}(\widehat{\delta})$. The interval estimates δ and remains centered at $\widehat{\delta} = 4$.

Problem 8: Does randomly assigned tutoring improve scores?

Students are individually randomized into independent tutoring (T) and control (C) groups, with 100 students in each. Treat observations as i.i.d. within each group. There is full take-up, no spillover, and no differential attrition.

	Tutoring	Control
Posttest mean (SD)	74 (10)	70 (10)
Baseline mean (SD)	69.5 (10)	70 (10)

Use large-sample normal inference: $z_{.975} = 1.96$, $\Phi(2.828) \approx .99766$, and $\Phi(.3536) \approx .6382$.

1. Estimate the posttest mean difference and its standard error.
2. Test no posttest effect at 5%. Find the two-sided p-value and 95% confidence interval.
3. Test equality of baseline means at 5%. What does the result establish?
4. Under the stated design, how can we interpret the posttest difference?

Solution 8: Estimating and testing the tutoring effect

1. For the observed posttest difference $\widehat{\Delta} = \bar{y}_T - \bar{y}_C$,

$$\widehat{\Delta} = 74 - 70 = 4 \text{ points}, \qquad \widehat{\text{Var}}(\widehat{\Delta}) = \frac{s_T^2}{n_T} + \frac{s_C^2}{n_C}.$$

The groups are independent, so

$$SE(\widehat{\Delta}) = \sqrt{\frac{s_T^2}{n_T} + \frac{s_C^2}{n_C}} = \sqrt{\frac{10^2}{100} + \frac{10^2}{100}} = \sqrt{2} \approx 1.414 \text{ points}.$$

2. Test $H_0 : \Delta = 0$ against $H_1 : \Delta \neq 0$:

$$t = \frac{\widehat{\Delta} - 0}{SE(\widehat{\Delta})} = \frac{4}{\sqrt{2}} \approx 2.828, \qquad p \approx 2[1 - \Phi(|t|)] \approx .0047.$$

Since $|t| > 1.96$ and $p < .05$, reject H_0. The 95% confidence interval is

$$\widehat{\Delta} \pm 1.96\, SE(\widehat{\Delta}) = 4 \pm 1.96\sqrt{2} \approx \boxed{[1.228,\ 6.772]} \text{ points}.$$

Solution 8: Baseline balance and a causal interpretation

3. Let Δ_B denote the population baseline mean difference. For $H_0 : \Delta_B = 0$ against $H_1 : \Delta_B \neq 0$,

$$\widehat{\Delta}_B = 69.5 - 70 = -.5, \qquad SE(\widehat{\Delta}_B) = \sqrt{\frac{10^2}{100} + \frac{10^2}{100}} = \sqrt{2}.$$

$$t_B = \frac{\widehat{\Delta}_B}{SE(\widehat{\Delta}_B)} = \frac{-.5}{\sqrt{2}} \approx -.3536, \qquad p_B \approx 2[1 - \Phi(.3536)] \approx 2(1 - .6382) = .7236.$$

Fail to reject at 5%. The sample gives little evidence of a baseline mean difference; this does not establish exact equality or prove that assignment was random.

4. Random assignment, full take-up, no spillovers, and no differential attrition allow the posttest difference to estimate the average causal effect of tutoring for the study population: an estimated increase of 4 points.

The design supports this interpretation. The baseline test is a diagnostic, not a proof of the design.

Problem 9: What do the school score summaries describe?

Nine ordered school scores are

$$60,\quad 65,\quad 70,\quad 70,\quad 75,\quad 80,\quad 80,\quad 85,\quad 90.$$

1. Find the sample mean, median, minimum, and maximum.
2. Define each quartile as the median of its half of the ordered data, excluding the overall median. Find Q_1, Q_3, and the interquartile range.
3. Compute the sample variance and standard deviation. Show the denominator used in the variance.
4. A *separate, larger district sample* reports skewness .37 and ordinary kurtosis 3.15 (not excess kurtosis). Interpret these summaries using the normal benchmark.

Solution 9: Location, quartiles, and spread

1. The sample mean is

$$\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i = \frac{675}{9} = 75 \text{ points.}$$

The median is the fifth ordered score, 75. The minimum is 60 and the maximum is 90.

2. Exclude the overall median before splitting the observations:

$$\underbrace{60,\ 65,\ 70,\ 70}_{\text{lower half}},\qquad 75,\qquad \underbrace{80,\ 80,\ 85,\ 90}_{\text{upper half}}.$$

$$Q_1 = \frac{65+70}{2} = 67.5, \qquad Q_3 = \frac{80+85}{2} = 82.5.$$

$$\text{IQR} = Q_3 - Q_1 = 82.5 - 67.5 = \boxed{15 \text{ points}}.$$

Solution 9: Variance and distribution shape (continued)

3. The sample variance uses the divisor $n - 1$:

$$\sum_{i=1}^{9}(y_i - \bar{y})^2 = 225 + 100 + 25 + 25 + 0 + 25 + 25 + 100 + 225 = 750,$$

$$s_y^2 = \frac{\sum_{i=1}^{n}(y_i - \bar{y})^2}{n-1} = \frac{750}{8} = 93.75 \text{ points}^2,$$

$$s_y = \sqrt{s_y^2} = \sqrt{93.75} \approx 9.682 \text{ points}.$$

4. For the *separate, larger sample*, skewness $.37 > 0$ indicates modest asymmetry toward the right.

Ordinary kurtosis 3.15 is slightly above the normal benchmark of 3: the standardized fourth moment is slightly larger.

These summaries do not determine exact tail probabilities. They are not the skewness and kurtosis of the nine listed scores.

Problem 10: Do study hours and test scores move together?

Five students provide paired observations of weekly study hours X and test scores Y:

Student	1	2	3	4	5
Hours x_i	1	2	3	4	5
Score y_i	60	70	65	80	75

1. Find $\bar{x}$ and $\bar{y}$. Tabulate the centered observations and their cross-products.
2. Compute the sample covariance and state its units.
3. Compute both sample standard deviations and the sample correlation. What association does it describe?
4. Convert hours to minutes, $X^* = 60X$. Find the covariance with Y and the correlation. Explain the change.

Solution 10: Centered observations and sample covariance

1. The sample means are

$$\bar{x} = \frac{1}{5}\sum_{i=1}^{5} x_i = \frac{15}{5} = 3 \text{ hours}, \qquad \bar{y} = \frac{1}{5}\sum_{i=1}^{5} y_i = \frac{350}{5} = 70 \text{ points}.$$

Student	1	2	3	4	5	Sum
$x_i - \bar{x}$	-2	-1	0	1	2	0
$y_i - \bar{y}$	-10	0	-5	10	5	0
$(x_i - \bar{x})(y_i - \bar{y})$	20	0	0	10	10	40

2. The sample covariance is

$$s_{XY} = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y}) = \frac{40}{5-1} = \boxed{10 \text{ hour-points}}.$$

Observations with higher study hours tend to have higher scores in this sample.

Solution 10: Correlation and a change of units (continued)

3. For either variable V, $s_V^2 = \sum_i (v_i - \bar{v})^2/(n-1)$. Thus

$$s_X = \sqrt{\frac{4+1+0+1+4}{4}} = \sqrt{2.5} \approx 1.581 \text{ hours},$$

$$s_Y = \sqrt{\frac{100+0+25+100+25}{4}} = \sqrt{62.5} \approx 7.906 \text{ points}.$$

$$r_{XY} = \frac{s_{XY}}{s_X s_Y} = \frac{10}{\sqrt{2.5}\sqrt{62.5}} = \frac{10}{12.5} = \boxed{.8}.$$

This is a positive linear association; correlation alone does not establish a causal effect.

4. A positive change of units gives

$$s_{X^* Y} = 60 s_{XY} = 600 \text{ minute-points}, \qquad s_{X^*} = 60 s_X,$$

$$r_{X^* Y} = \frac{60 s_{XY}}{(60 s_X) s_Y} = r_{XY} = .8.$$

Covariance depends on units; the common scale factor cancels in correlation.

Problem 11: Are scores different across disadvantaged districts?

Districts are classified by their share of students receiving subsidized meals. The samples are independent, with i.i.d. districts within each group:

Share receiving subsidized meals	n	Mean $\bar{y}$	SD s
Low (L)	200	780	50
High (H)	200	720	50

Use large-sample normal inference, $z_{.975} = 1.96$ and $\Phi(3.291) \approx .9995$.

1. Estimate $\Delta = \mu_L - \mu_H$ and its standard error.
2. Test $H_0 : \Delta = 0$ against $H_1 : \Delta \neq 0$ at 5%. Bound the two-sided p-value using the supplied normal probability.
3. Construct a 95% confidence interval for the mean difference.
4. Does this estimate the causal effect of subsidized meals on test scores? Explain.

Solution 11: The gap and its statistical significance

1. The estimated mean difference is

$$\widehat{\Delta} = \bar{y}_L - \bar{y}_H = 780 - 720 = 60 \text{ points.}$$

For independent groups,

$$SE(\widehat{\Delta}) = \sqrt{\frac{s_L^2}{n_L} + \frac{s_H^2}{n_H}}$$
$$= \sqrt{\frac{50^2}{200} + \frac{50^2}{200}} = \sqrt{12.5 + 12.5} = \boxed{5 \text{ points}}.$$

2. Under $H_0 : \Delta = 0$,

$$t = \frac{\widehat{\Delta} - 0}{SE(\widehat{\Delta})} = \frac{60}{5} = 12.$$

Since $|t| > 3.291$, the two-sided normal p-value satisfies

$$p \approx 2[1 - \Phi(12)] < 2[1 - \Phi(3.291)] \approx .001.$$

Reject H_0 at 5%. The p-value is very small; it is not exactly zero.

Solution 11: A confidence interval and its interpretation

3. The large-sample 95% confidence interval is

$$\widehat{\Delta} \pm 1.96\, SE(\widehat{\Delta}) = 60 \pm 1.96(5)$$
$$= 60 \pm 9.8 = \boxed{[50.2,\ 69.8]} \text{ points.}$$

The entire interval is positive, agreeing with rejection of the zero-difference null in the two-sided 5% test.

4. The low-share districts have higher mean scores in these observational data. The difference does not identify a causal effect of subsidized meals.

Districts may differ in income, family background, school resources, and other determinants of scores. Statistical significance does not remove these confounding differences.